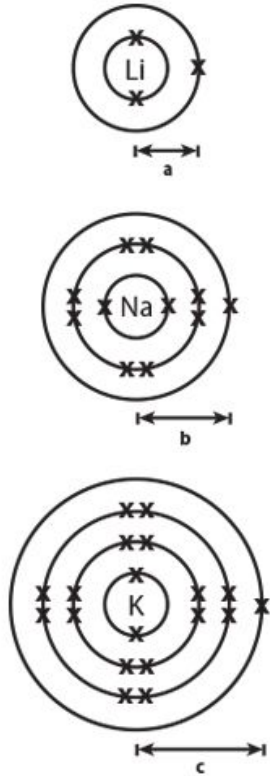


Trends in groups of Periodic table



Group 1

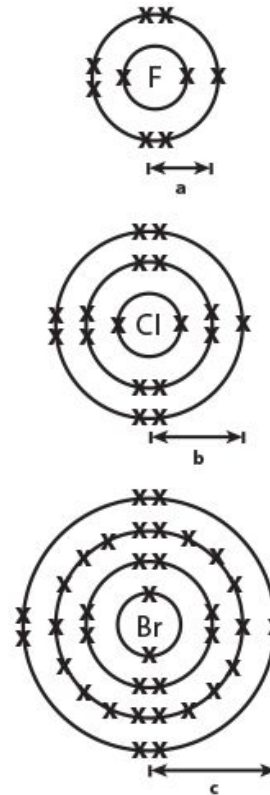
Li
Na
K
Rb
Cs
Fr

Reactivity increases down the group

This is because... the outer electron is getting further away from the nucleus

This means the attraction is lower

The outer electron is easier to lose



Group 7

F
Cl
Br
I
At
Ts

Reactivity increases up the group

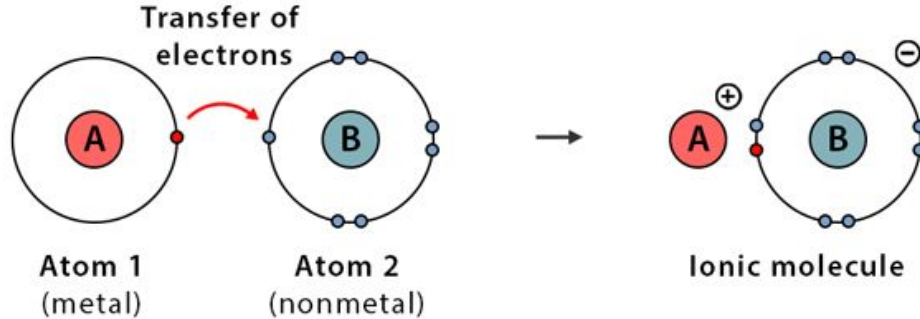
This is because... the outer electron is getting closer to the nucleus

This means the attraction is higher

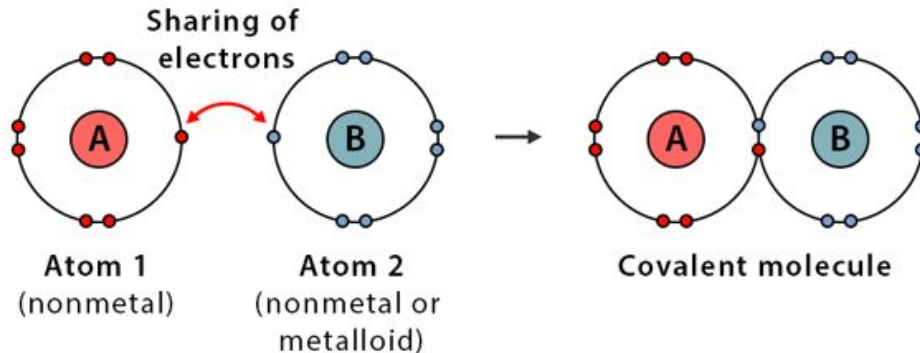
It is easier to gain an electron

Ionic and covalent bonds

Ionic Bond



Covalent Bond

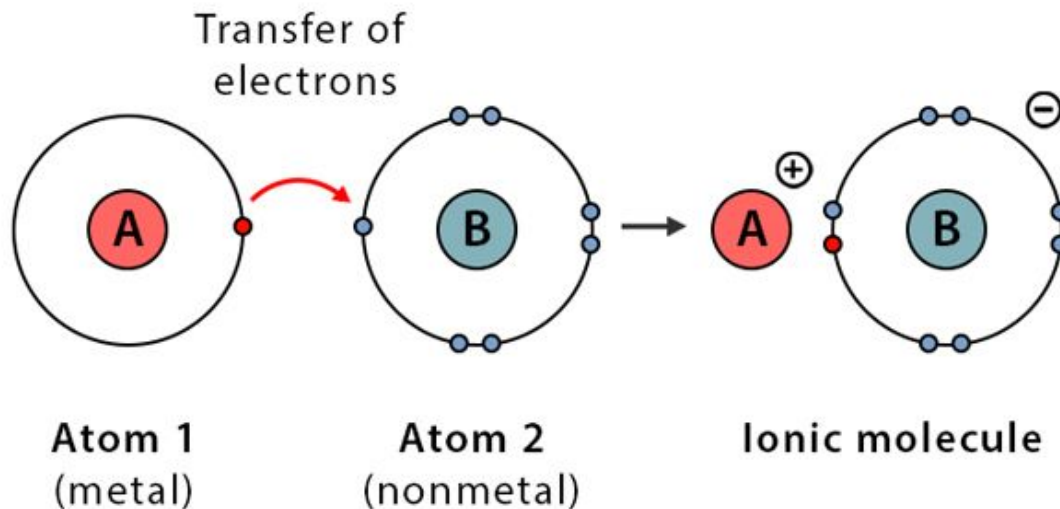


Atoms are more **stable** when the outermost electron shell is completely full of electrons.

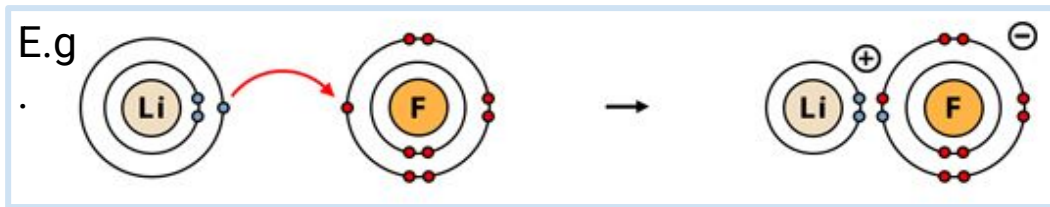
Elements react to **form compounds** to do this. The elements are held together by chemical bonds by:

1. Atoms losing or gaining electrons (**ionic bond**)
2. Atoms sharing electrons (**covalent bond**)

Ionic bonds

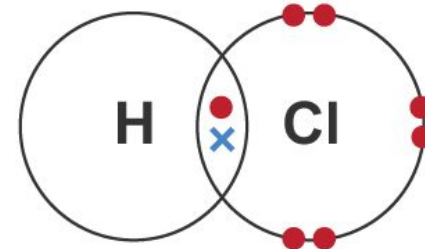
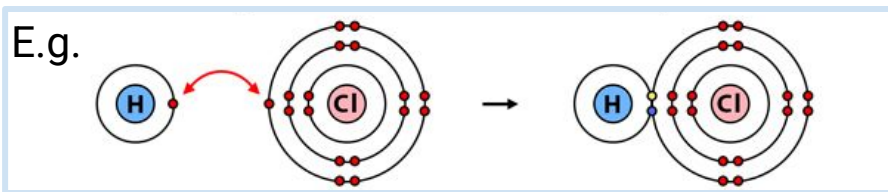
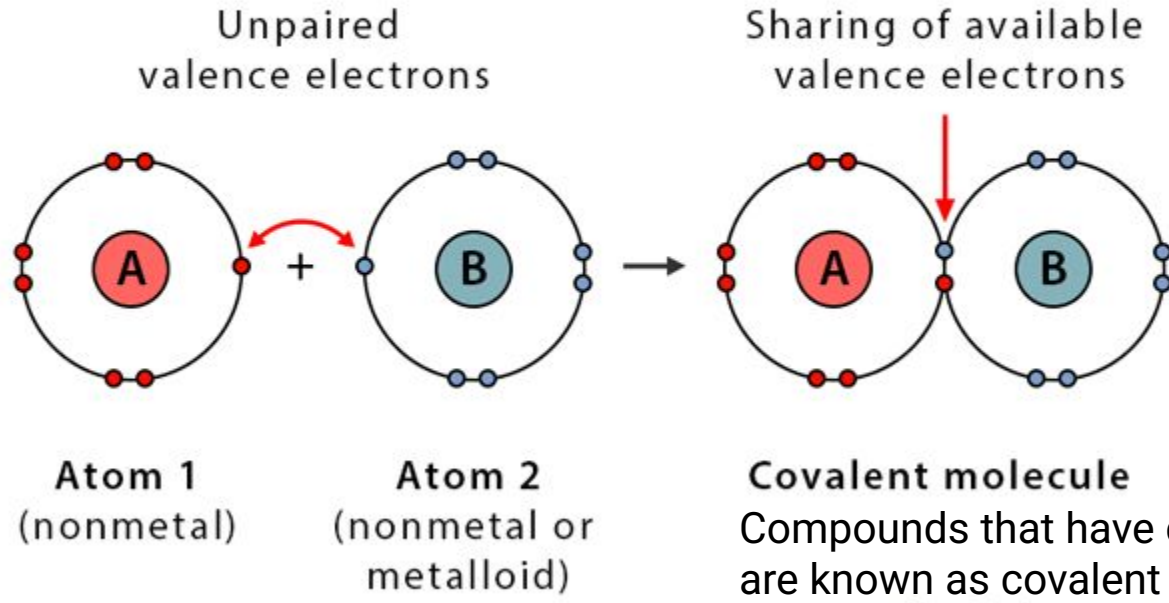


Li
Lithium [2,1]
F
Fluorine [2,7]

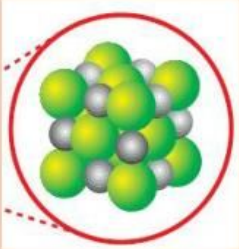
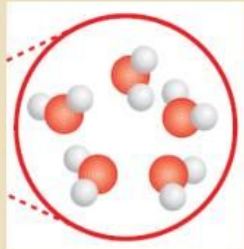
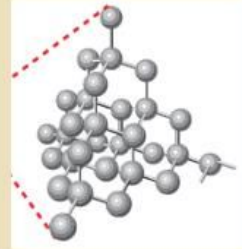
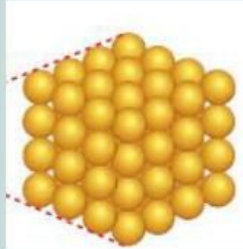


Li⁺
Lithium ion [2]⁺
F⁻
Fluoride ion [2,8]⁻

Covalent bonds

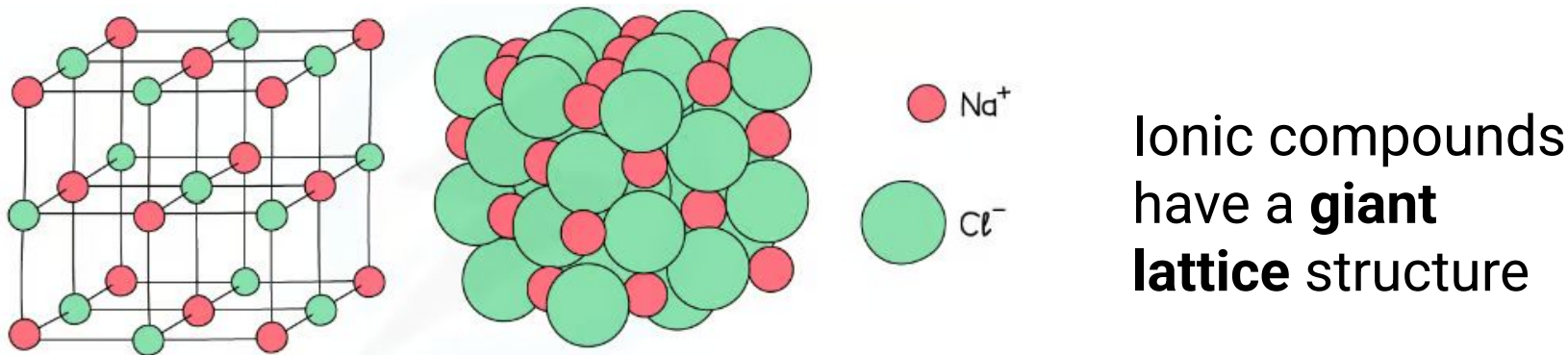


Simple and giant structures

Bonding	Ionic	Covalent		Metallic
Structure	Giant ionic	Simple molecular	Giant covalent	Giant Metallic
Example	Sodium chloride 	Water 	Diamond 	Iron 

Bonding and structure explains the properties of a substance!

Giant ionic structures



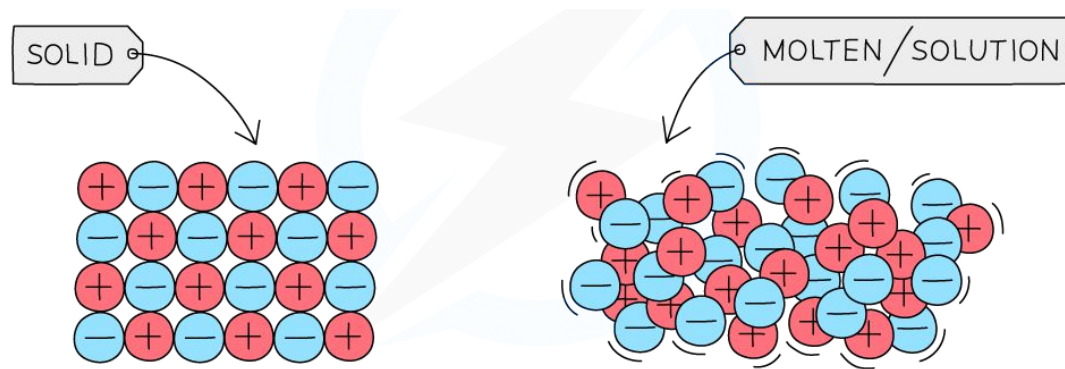
The lattice consists of a regular arrangement of **alternating positive and negative ions** in which the ions are tightly packed together.

Strong **electrostatic forces of attraction** are present between oppositely charged ions, holding the lattice together.

Giant ionic structures

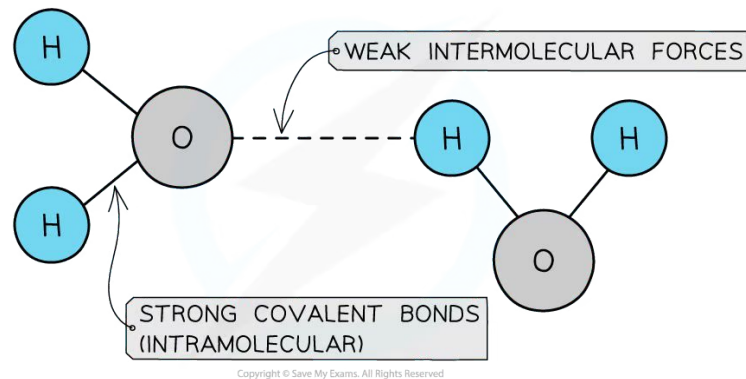
Ionic compounds have **high melting and boiling points** as the forces need lots of energy to overcome them.

Ionic compounds are **poor conductors in the solid state** as the ions are in fixed positions in the lattice so they are unable to move and carry a charge. However, when the ionic compound is **melted or dissolved in water, they are good conductors** of electricity as the ions are able to move and carry a charge.



Simple molecular structures

- Small molecules are compounds made up of molecules that contain just **a few atoms covalently bonded together**
- They have **low melting and boiling points** so covalent compounds are usually liquids or gases at room temperature
- Small molecules have **poor electrical conductivity** as there are **no free ions** or electrons to carry the charge.



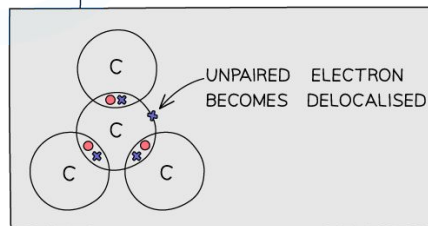
Giant covalent structures

Diamond and graphite are **allotropes** of carbon - different molecular arrangements of the same element in the same state.

GRAPHITE

EACH CARBON ATOM
FORMS 3 COVALENT
BONDS WITH 3
OTHER CARBON ATOMS

WEAK
INTERMOLECULAR
FORCES OF
ATTRACTION



DIAMOND

GIANT
COVALENT
STRUCTURE

CARBON
ATOMS

COVALENT BOND
(SHARED PAIR
OF ELECTRONS)
BETWEEN 2 CARBON
ATOMS

EACH CARBON
ATOM FORMS
FOUR COVALENT
BONDS WITH
OTHER CARBON
ATOMS

(NOT ALL BONDS
ARE SHOWN)

Have a go at Workbook questions - see Classroom

Topic 2.4 Simple and giant structures

Exercise 2.4A Ionic or covalent bonds

- 1 Ionic bonding, because the particles are packed closely together in a lattice pattern.
- 2 Arrows should indicate the points where the large atom and the four small atoms in each molecule join. The label should read 'strong forces within each molecule'.
- 3 Substance B
- 4 Substance A
- 5 Ionic
- 6 covalent
- 7 Any correct example, such as graphite or silicon dioxide.

Exercise 2.4B Properties of ionic and covalent substances

- 1 Its melting and boiling points are high.
- 2 a gas
- 3 potassium chloride, calcium chloride and magnesium oxide
- 4 ammonia and bromine
- 5 Water; because it has a boiling point of 100°C and a melting point of 0°C .
- 6 bromine

- 7 Magnesium oxide has high melting and boiling points because it is an ionic compound. The ions have strong forces holding them together in a lattice pattern. This means that a lot of energy is needed to overcome these forces and make the magnesium oxide melt or boil.
- 8 Ammonia has low melting and boiling points because, although the forces inside the molecules are strong, the forces between the molecules are weak and less energy is needed to overcome these intermolecular forces and make the ammonia melt or boil.

Exercise 2.4C Giant structures of carbon

- 1 diamond (left) and graphite (right)
- 2 The atoms of carbon in graphite are arranged in layers. The bonds between the carbon atoms in the layers are strong (each carbon atom bonds with three other atoms) but the bonds between the layers are weak. This means that the layers can slide over one another and the surface is soft and comes away. For example, when you write with a graphite pencil.
- 3 The atoms of carbon in diamond are arranged in a rigid, giant three-dimensional structure or lattice. This means that there are strong bonds throughout the whole structure and this is what makes diamond so hard.